



Innovation Web Application Project

PROJECT DESCRIPTION

Create a small full-stack web application that meaningfully improves or innovates an existing process, service, or system. The project must solve a real user problem and demonstrate a clear improvement over the current way of doing the task.

1. Project Overview

This is a two-week innovation project followed by a Week 3 awarding and recognition activity. Identify one real problem, build a focused solution, test it, and defend why it helps its target user.

CORE PRINCIPLE

Build one useful idea well. A smaller system with a clear purpose, reliable workflow, and meaningful innovation is stronger than a large system with many unfinished features.

Project Goal

- Apply Vue.js in a working web application.
- Use Tailwind CSS for a clear, usable, responsive interface.
- Connect the interface to persistent data through an approved backend/API/database.
- Improve an existing process for a specific target user.
- Show testing, debugging, technical understanding, and clear defense of design decisions.

Grading Structure

Assessment	When	Maximum	Important Note
Combined Laboratory 1 & 2	Week 1	50 points	70–80% checkpoint using the current Lab 2 criteria
Final Defense	Week 2	50 points	Final defense and live system test
Awarding & Recognition	Week 3	Not graded	1 Best System per section + 1 Overall Best System

PROJECT GRADE

The score earned in the Final Defense will also be recorded as the student/group Project grade. Example: a Final Defense score of 44/50 will also be the Project grade of 44/50.



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2. Required Project Scope

To finish within the two-week project schedule, follow the limits below unless the instructor approves an exception.

Scope Rule	Expected Limit
Problem	One primary problem
Target user	One clearly defined main user or user group
Main workflow	One complete end-to-end workflow
Core features	Approximately three essential features
Innovation	At least one clearly demonstrated improvement
Group size	Maximum of 4 members and minimum of 3 members per group
Development period	Two project weeks + Week 3 awarding and recognition

Avoid oversized project titles and scopes. Choose one focused part of a larger domain; do not recreate an entire hospital, school, store, hotel, LMS, or office system.

Too Broad	Better Focused Scope
Hospital Management System	Clinic Queue Prioritization System
Learning Management System	Assignment Progress and Deadline-Risk Tracker
Inventory Management System	Low-Stock Prioritization and Reorder Assistant
School Management System	Student Consultation Scheduling and Queue System

3. What Counts as Innovation?

Innovation means making an existing process meaningfully better for a specific user.

REQUIRED INNOVATION STATEMENT

"Existing systems/processes _____. Our system improves this by _____, which helps users _____."

Possible Forms of Innovation

- Automation or fewer steps.
- Faster decisions, prioritization, or smart sorting.



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- Error prevention and validation.
- Better accessibility or usability.
- Clearer visualization, summaries, or recommendations.

Important: Adding login, CRUD pages, a dashboard, or a new color scheme alone is not innovation; the improvement must change how users solve the problem.

Measurable Improvement

In the final defense, show what improved: fewer steps, less manual work, faster access, fewer errors, or clearer decisions.

4. Minimum Technical Requirements

The final application must meet these minimum requirements:

- Vue.js is used as the primary front-end framework.
- Tailwind CSS is used for the application interface and responsive layout.
- The application contains a complete main user workflow, not only disconnected pages.
- User input is validated and common invalid actions are handled appropriately.
- Important data persists using an appropriate backend, API, database, or storage approach approved for the course.
- CRUD operations are implemented where they are naturally required by the system.
- The proposed innovative feature is functional and visible in the final demonstration.
- The interface provides clear feedback for successful actions, errors, empty states, and important system states.
- The system uses realistic test data and must not rely on a purely hard-coded/fake demonstration for its main workflow.

RESPONSIBILITY FOR CODE

You may use references, documentation, libraries, or AI-assisted development as allowed by the instructor. However, you are responsible for every part of the submitted system. During the defense, you must be able to explain how important code works and make reasonable small changes when asked.

5. Project Timeline and Recognition Schedule

Week	Lecture Focus	Project Activity	Expected Result
1	Vue/Tailwind, architecture, forms, state, and data/API integration	LAB 1 & 2: Combined Checkpoint and Mock Defense	Core system at 70–80% completion using the current Lab 2 criteria
2	Testing, debugging, UX, deployment, and defense prep	FINAL DEFENSE	Final system defended through live demo and Q&A
3	Recognition and reflection	AWARDING & RECOGNITION	1 Best System per section + 1 Overall Best System



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6. Laboratory 1 & 2 — Combined Project Checkpoint & Mock Defense

Maximum score: 50 points

Purpose: Present the working system, test its main workflow, find weaknesses, and record fixes. Target about 70–80% completion. Proposal and design materials support the review; there is no separate Lab 1 proposal score.

Supporting Proposal and Design Evidence

1. **Proposed System Title:** Specific title for the focused problem or improvement.
2. **Target User(s):** Who will use or benefit from the system.
3. **Problem Statement:** What problem exists and why it matters.
4. **Current Process / Existing Approach:** How the task is done now and where it fails.
5. **Proposed Solution:** What the application will do.
6. **Innovation Statement:** Complete the required innovation sentence.
7. **Core Features:** List about three must-have features; mark extras as optional.
8. **Scope and Limitations:** What the project includes and excludes.

Combined Lab 1 & 2 Rubric

Criteria	What will be evaluated	Points
System Completeness & Core Workflow	The main end-to-end workflow is functional (~70–80% complete) with connected forms, navigation, persistent data operations, input validation, and responsive usability.	20
Innovative Feature Implementation	The proposed innovation is implemented and functional enough to evaluate, demonstrating real workflow improvement rather than generic CRUD/login alone.	15
Presentation, Defense & Testing Log	Clear system walkthrough; ability to explain progress and system behavior; documented Bug/Testing Log with attempted fixes; effective response to instructor stress tests.	15
TOTAL		50



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7. Week 2 — Final Innovation Defense

Maximum score: 50 points

Important: The Final Defense score is also the Project grade.

Purpose: Prove the application works, the innovation is useful, and each student understands the submission.

Final Defense / Project Rubric

Criteria	What will be evaluated	Points
System Completeness & Functionality	Complete, robust end-to-end workflow; persistent data operations; polished and responsive interface; reliable error handling under normal and edge conditions.	20
Innovation & Real-World Usefulness	The innovative feature is fully functional, delivers measurable improvement, solves the target user's problem effectively, and provides clear practical value.	15
Presentation, Defense & System Ownership	Articulate the live demonstration following the 5-step flow; explain the system architecture and data flow; confidently handle unrehearsed instructor tests and technical Q&A.	15
TOTAL		50

8. Week 3 — Awarding & Recognition

- **Best System Award:** one award for the best system in each section.
- **Overall Best System Award:** one award selected from the best systems across all sections combined.
- **Recognition basis:** innovation, real-world usefulness, system completeness, presentation, and defense performance. Week 3 is not a separate graded meeting.

9. Final Reminder

WHAT YOU ARE DEFENDING

You are not defending the size of your project. You are defending that you identified a real problem, created a useful improvement, implemented it correctly, tested it, and understand how your system works.

Discover → Design → Build → Test → Defend